Chapter 6 solutions

CH06-E1

1.

$$p_A(\lambda) = (3-\lambda)^2 - 1 = (\lambda - 4)(\lambda - 2).$$

Eigenvalues $4, 2$.

2. For $\lambda = 4$, the equations give $x_1 = x_2$, so

$$E_4 = \text{span}\{(1,1)^T\}.$$

For $\lambda = 2$, they give $x_1 + x_2 = 0$, so

$$E_2 = \text{span}\{(1,-1)^T\}.$$

Both geometric multiplicities are 1.

3.

$$u_0 = (2,0)^T = (1,1)^T + (1,-1)^T.$$

$$u_k = 4^k(1,1)^T + 2^k(1,-1)^T = \begin{bmatrix} 4^k + 2^k \\ 4^k - 2^k \end{bmatrix}.$$

CH06-E2

1. Trace 5, det 6:

$$p_A(\lambda) = (\lambda - 3)(\lambda - 2).$$

For eigenvalues 3 and 2, take $x_1 = (2,1)^T$ and $x_2 = (1,1)^T$, respectively.

2.

$$X = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad X^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

$$A = X \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} X^{-1}.$$

3.

$$A^5 = X \begin{bmatrix} 243 & 0 \\ 0 & 32 \end{bmatrix} X^{-1}.$$

Multiplying gives

$$\begin{bmatrix} 486 & 32 \\ 243 & 32 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 454 & -422 \\ 211 & -179 \end{bmatrix}.$$

CH06-E3

1. Both triangular diagonal entries are 2.

$$p_A(\lambda) = (\lambda - 2)^2.$$

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Chapter 6 solutions (continued)

Algebraic multiplicity 2.

$$(A - 2I)x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

Only $x_2 = 0$ is imposed, so

$$E_2 = \text{span}\{(1,0)^T\}, \qquad g_2 = 1.$$

2. Only one eigenvector direction exists, so diagonalization is impossible.

3.

$$A = 2I + N, \qquad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N^2 = 0.$$

Binomial:

$$A^k = 2^k I + k2^{k-1} N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}.$$

CH06-E4

1. Matrix

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

has eigenvalues 3 and 1 with orthonormal eigenvectors

$$q_1 = \frac{1}{\sqrt{2}}(1,-1)^T,$$
$$q_2 = \frac{1}{\sqrt{2}}(1,1)^T.$$

2.

$$u(0) = (1,3)^T$$
$$q_1^T u(0) = -\sqrt{2}, \qquad q_2^T u(0) = 2\sqrt{2}.$$

3.

$$u(t) = -\sqrt{2}e^{3t} q_1 + 2\sqrt{2}e^{t} q_2.$$

Expanding gives

$$u(t) = \begin{bmatrix} -e^{3t} + 2e^{t} \\ e^{3t} + 2e^{t} \end{bmatrix}.$$

Both eigenvalues are positive. For these initial data, the eigenvalue-3 mode is nonzero and dominates the growth.

Chapter 6 solutions (continued)

CH06-E5

1.

$$p(\lambda) = \lambda^2 + 1, \qquad \lambda = \pm i.$$

For $\lambda = i$, choose $(1, -i)^T$; for $\lambda = -i$, choose its conjugate $(1, i)^T$.

2. Real solution：

$$u(t) = \begin{bmatrix} c_1 \cos t - c_2 \sin t \\ c_1 \sin t + c_2 \cos t \end{bmatrix}.$$

$u(0) = (c_1, c_2)^T$ 。

3.

$$\frac{d}{dt}(x^2 + y^2) = 2x(-y) + 2y(x) = 0.$$

Thus the orbit lies on a circle.

4. Matrix

$$\begin{bmatrix} \alpha & -1 \\ 1 & \alpha \end{bmatrix}$$

The eigenvalues are $\alpha \pm i$. For $\alpha < 0$, solutions decay and zero is asymptotically stable. At $\alpha = 0$, bounded rotation gives Lyapunov stability but not asymptotic stability. For $\alpha > 0$, nonzero solutions spiral outward and zero is unstable.

CH06-E6

1. Direct transposition gives $S^T = S$, and the energy is

$$2(x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_2x_3).$$

a symmetric quadratic form.

2. From $Sx = \lambda x$, multiply by x^*.

$$x^*Sx = \lambda x^*x.$$

The left side equals its conjugate and is real. Since $x^*x > 0$, the eigenvalue is real.

3. $Sx = \lambda x$, $Sy = \mu y$:

$$\lambda x^T y = (Sx)^T y = x^T Sy = \mu x^T y.$$

Distinct eigenvalues force $x \perp y$ for real eigenvectors, or $x^*y = 0$ for complex choices.

4. There exist an orthonormal eigenbasis Q and a real diagonal Λ such that

$$S = Q\Lambda Q^T.$$

Chapter 6 solutions (continued)

Each vector must belong to the appropriate eigenspace. Orthonormal choices can vary within repeated eigenspaces.

CH06-E7

1. Characteristic:

$$p_S(\lambda) = (\lambda - 1)(\lambda - 3),$$

Both eigenvalues, 3 and 1, are positive.

Leading minors:

$$\Delta_1 = 2 > 0, \qquad \Delta_2 = 3 > 0.$$

Energy:

$$x^T S x = x_1^2 + (x_1 - x_2)^2 + x_2^2 > 0$$

for every nonzero x.

2. Spectral coordinates:

$$3y_1^2 + y_2^2 = 1.$$

$|y_1|$ axis length $1/\sqrt{3}$, $|y_2|$ axis length 1.

3. On the unit sphere, the maximum energy is 3 and the minimum is 1, attained at the corresponding unit eigenvectors.

CH06-E8

1. The upper-triangular diagonal entries are all -1.

$$\lambda = -1, -1, -1.$$

Trace -3 › det -1 ∘

2. Let

$$N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad N^3 = 0.$$

$$e^{At} = e^{-t} \begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}.$$

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Chapter 6 solutions (continued)

3. All real parts are $-1 < 0$, so zero is asymptotically stable and $e^{At} \to 0$.
4. In general, asymptotic stability does not imply monotone decay. This matrix does contract in the 2-norm, since the largest eigenvalue of $(A + A^T)/2$ is $-1 + \sqrt{2}/2 < 0$, giving $d\|u\|^2/dt = u^T(A + A^T)u < 0$. A true counterexample is $B = \begin{bmatrix} -1 & 4 \\ 0 & -1 \end{bmatrix}$. Its eigenvalues are both -1, but at $u = (1, 1)^T$, $u^T(B + B^T)u = 4 > 0$. Asymptotic stability ensures eventual decay; a polynomial factor alone does not prove transient growth.

CH06-E9

1. A, B invertible:

$$BA = A^{-1}(AB)A.$$

Thus BA is similar to AB, and they share eigenvalues.

2. A diagonalizable A with all eigenvalues zero annihilates every eigenbasis vector.

$$Av_i = 0.$$

Since these vectors span, $A = 0$.

3. A counterexample is

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

It has only eigenvalue zero but is nonzero and defective.

CH06-E10

Since d_5=10 is the full dimension, all later nullities equal 10. Increments for k=1 through 6 are (3,3,2,1,1,0), counting blocks of length at least k. Their adjacent differences are (0,1,1,0,1), so there is one block of each length 2,3,5:

$$J = \mathrm{diag}(J_5(0), J_3(0), J_2(0)).$$

The dimensions add to 10. Rank-nullity gives rank N=10−3=7 and rank N^2=10−6=4. Since d_4=9<10=d_5, $N^4 \neq 0$ and N^5=0, so the least m is 5. The value d_1=3 counts chain endpoints and hence blocks, not maximum chain length.

CH06-E11

Write B=−I+4N, where $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and N^2=0. Then

$$e^{Bt} = e^{-t}(I + 4tN), \qquad u(t) = e^{-t}(1 + 4t, 1)^T/\sqrt{2}.$$

Chapter 6 solutions (continued)

For arbitrary initial data $(a,b)^T$ the solution is $e^{-t}(a+4tb,b)^T$. Both e^{-t} and te^{-t} tend to zero. The evolution operator is also bounded on $t \geq 0$, so the zero solution is asymptotically stable.

Set $h(t) = \|u(t)\|_2^2 = e^{-2t}[(1+4t)^2+1]/2$. Its derivative is $h'(t) = e^{-2t}(4z - z^2 - 1)$, with $z = 1+4t \geq 1$. The only admissible root is $z = 2+\sqrt{3}$. The derivative is positive before it and negative after it, so

$$t_* = (1+\sqrt{3})/4, \quad \max_{t \geq 0} \|u(t)\|_2 = e^{-(1+\sqrt{3})/4}\sqrt{4+2\sqrt{3}} \approx 1.379.$$

For C the same initial state gives $e^{-t}u_0$, whose norm decreases from 1. For B, $H = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$ satisfies $Hu_0 = u_0$, so $h'(0) = 2$. But initial state e_1 gives $e^{-t}e_1$ and decays monotonically.

Part 2 fixes one initial direction. The operator norm $\|e^{Bt}\|_2 = \max_{\|v\|_2=1} \|e^{Bt}v\|_2$ optimizes that direction separately at each t. Our curve is only a lower bound on the operator norm.

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Chapter 6 solutions (continued)

Verification code appendix: eigen_dynamics.py

```
"""Check eigen-decompositions, discrete powers, ODE and definiteness examples."""
from math import exp, sqrt
Vec = tuple[float, ...]
Mat = tuple[tuple[float, ...], ...]

def matvec(matrix: Mat, vector: Vec) -> Vec:
    return tuple(
        sum(matrix[i][j] * vector[j] for j in range(len(vector)))
        for i in range(len(matrix))
    )

def matmul(left: Mat, right: Mat) -> Mat:
    return tuple(
        tuple(
            sum(left[i][k] * right[k][j] for k in range(len(right)))
            for j in range(len(right[0]))
        )
        for i in range(len(left))
    )

def residual(matrix: Mat, vector: Vec, value: float) -> float:
    image = matvec(matrix, vector)
    return sqrt(sum((image[i] - value * vector[i]) ** 2 for i in range(len(vector))))

def matrix_power(matrix: Mat, exponent: int) -> Mat:
    result = tuple(
        tuple(1.0 if i == j else 0.0 for j in range(len(matrix)))
        for i in range(len(matrix))
    )
    base = matrix
    while exponent:
        if exponent & 1:
            result = matmul(result, base)
        base = matmul(base, base)
        exponent >>= 1
    return result
a = ((3.0, 1.0), (1.0, 3.0))
x1 = (1 / sqrt(2), 1 / sqrt(2))
x2 = (1 / sqrt(2), -1 / sqrt(2))
assert residual(a, x1, 4.0) < 1e-12
assert residual(a, x2, 2.0) < 1e-12
assert abs(x1[0] * x2[0] + x1[1] * x2[1]) < 1e-12
u0 = (2.0, 0.0)
```

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Chapter 6 solutions (continued)

```
for k in range(8):
    closed = (
        4**k + 2**k,
        4**k - 2**k,
    )
    iterative = matvec(matrix_power(a, k), u0)
    assert closed == iterative
defective = ((2.0, 1.0), (0.0, 2.0))
assert matrix_power(defective, 5) == (
    (32.0, 80.0),
    (0.0, 32.0),
)
t = 0.75
system = ((2.0, -1.0), (-1.0, 2.0))
closed_u = (
    -exp(3.0 * t) + 2.0 * exp(t),
    exp(3.0 * t) + 2.0 * exp(t),
)
steps = 200
dt = t / steps
u = (1.0, 3.0)
for _ in range(steps):
    k1 = matvec(system, u)
    k2 = matvec(system, tuple(u[i] + 0.5 * dt * k1[i] for i in range(2)))
    k3 = matvec(system, tuple(u[i] + 0.5 * dt * k2[i] for i in range(2)))
    k4 = matvec(system, tuple(u[i] + dt * k3[i] for i in range(2)))
    u = tuple(
        u[i] + dt * (k1[i] + 2.0 * k2[i] + 2.0 * k3[i] + k4[i]) / 6.0
        for i in range(2)
    )
assert max(abs(u[i] - closed_u[i]) for i in range(2)) < 1e-8
positive_definite = ((2.0, -1.0), (-1.0, 2.0))
for probe in ((1.0, 0.0), (0.0, 1.0), (1.0, -1.0), (2.0, 1.0)):
    energy = probe[0] * matvec(positive_definite, probe)[0] + probe[1] * matvec(
        positive_definite, probe
    )[1]
    assert energy > 0
print("A^100 =", matrix_power(a, 100))
print("defective A^5 =", matrix_power(defective, 5))
print("u(0.75) =", closed_u)
print("energy samples are positive")
```

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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